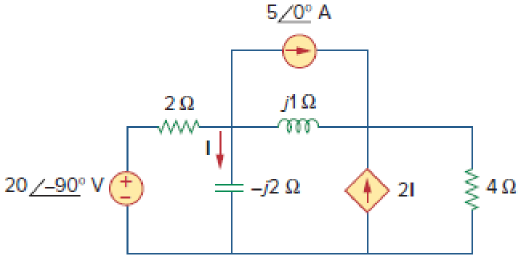
Chapter 10, Problem 15P

Problem

Solve for the current **I** in the circuit of Fig. 10.64 using nodal analysis.

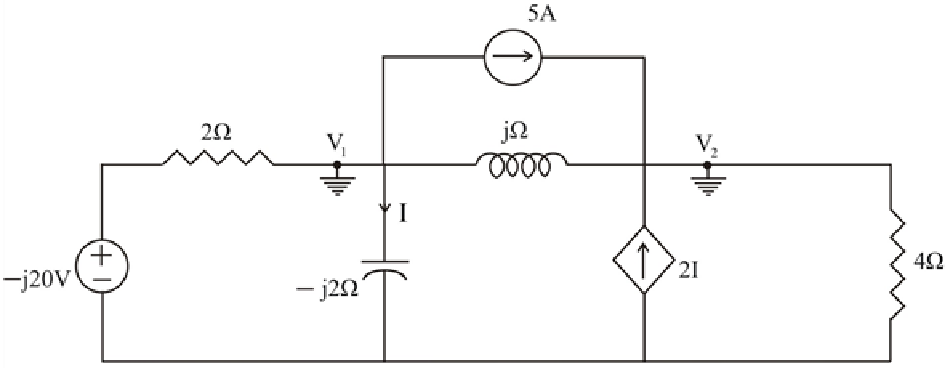
Figure 10.64:



Step-by-step solution

Step 1 of 5

We apply Nodal analysis to the circuit shown below.



Step 2 of 5

At node 1,

$$\frac{-j20-V_1}{2}=5+\frac{V_1}{-j2}+\frac{V_1-V_2}{j}$$

$$\frac{-j20-V_1}{2}=5+\frac{jV_1}{2}-j(V_1-V_2)$$

$$-j20-V_1=10+jV_1-j2(V_1-V_2)$$

$$-j20=10+V_1(j+1-j2)-j2(-V_2)$$

$$-j20=10+V_1(1-j)+j2V_2$$

$$-5-j10=(0.5-j0.5)V_1+jV_2 \quad\text{................ (1)}$$

Step 3 of 5

At node 2,

$$5+2I+\frac{V_1-V_2}{j}=\frac{V_2}{4}$$

Where,

$$I=\frac{V_1}{-j2}$$

Hence,

$$5+2\left(\frac{V_1}{-j2}\right)+\frac{V_1-V_2}{j}=\frac{V_2}{4}$$

$$5+jV_1-j(V_1-V_2)=\frac{V_2}{4}$$

$$20+j4V_1-j4(V_1-V_2)=V_2$$

$$20+V_1(j4)-j4V_1+j4V_2=V_2$$

$$20+j4V_2=V_2$$

$$20=V_2(1-j4)$$

$$V_2=\frac{20}{(1-j4)}$$ …………….. (2)

Step 4 of 5

Substituting (2) into (!),

$$-5-j10=(0.5-j0.5)V_1+j\left(\frac{20}{(1-j4)}\right)$$

$$-45+j10=(-1.5-j2.5)V_1+20j$$

$$-45-j10=(-1.5-j2.5)V_1$$

$$\frac{-45-j10}{(-1.5-j2.5)}=V_1$$

$$V_1=10.8823-j11.4706$$

Or,

$$V_1=15.811\angle-46.51^\circ\ \text{V}$$

Step 5 of 5

Therefore,

$$I=\frac{V_2}{-j2}$$

$$I=\frac{15.811\angle-46.51^\circ}{-j2}$$

$$I=\frac{15.811\angle-46.51^\circ}{2\angle-90^\circ}$$

$$\therefore \boxed{I=7.906\angle 43.49^\circ\ \text{A}}$$